

CSE2315 — Assignment 8

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March 26, 2026

Exercise 1

Consider the problem Hamiltonian Circuit (HAMC):

$\{(V, E) \mid \text{The graph } G = (V, E) \text{ contains a simple cycle } \pi \text{ such that } \forall v \in V : v \in \pi\}$

In a different notation we would write:

Name: Hamiltonian Circuit (HAMC)

Input: A graph $G = (V, E)$.

Question: Is there a simple cycle π in G such that all vertices $v \in V$ are in that cycle?

Now take the following reduction from HAMC to HAMP, given an instance $G = (V, E)$ of HAMC:

- I. Copy G and add 2 new nodes a, b to V .
- II. Pick some vertex $v_x \in V$: connect b to all neighbours of v_x and connect a to v_x .

Show that this reduction is not reliable.

Note: this can be quite challenging! Try and apply the reduction on some graphs and see what happens first, get a feeling for how the reduction influences the graphs.

Answer: The reduction is not reliable because the following does not hold: $\langle G \rangle \notin \text{HAMC} \rightarrow f(\langle G \rangle) \notin \text{HAMP}$. Take as a counterexample the graph $G = (\{v_1, v_2, v_3\}, \{\{v_1, v_2\}, \{v_2, v_3\}\})$. This graph does not have a Hamiltonian Circuit. However, if we take $f(\langle G \rangle)$ and let the v_x that is picked in step II be v_2 , then we obtain the new graph:

$$G' = (\{v_1, v_2, v_3, a, b\}, \{\{v_1, v_2\}, \{v_2, v_3\}, \{v_1, b\}, \{v_3, b\}, \{v_2, a\}\})$$

which has a Hamiltonian path (a, v_2, v_3, b, v_1)

Exercise 2

After many hours of hard work, Prof. P. E. Kwalsenpee has managed to create an algorithm TSP-Decider that can solve the decision variant of TSP in polynomial time. That is, given a set

$$S = \{s_i\}_{i=1}^n$$

of cities, a $n \times n$ intercity distance matrix $D = [d_{i,j}]$ where $d_{i,j} \in \mathbb{Z}^+$ and a positive integer k , the algorithm will accept if and only if there is a tour through all vertices in S such that the total cost of the tour is $\leq k$.

- (a) **Explain how we can also use TSP-Decoder to decide the Hamiltonian Circuit problem in polynomial time.**

We define a reduction $HAMC \leq_P TSP$ by using the following mapping function f :

$$\begin{aligned} f : \Sigma^* &\rightarrow \Sigma^* \\ f(\langle G = \{V, E\} \rangle) &= \langle S, D, k \rangle \end{aligned}$$

Where $S = V, k = |V|$ and $D = [d_{i,j}]$ where $d_{i,j} = 1$ if $\{v_i, v_j\} \in E$ or ∞ otherwise. f defines a valid Karp reduction as it is both efficient (computable in polynomial time) and reliable.

1. **Efficiency:** The function f needs to write 3 things on the tape: S, D and k :

- S is just a copy of V of the original graph, so it can be written in linear time
- The matrix D is $|V| \times |V|$, so the size is polynomial in the size of the input. And actually calculating each cell of the matrix is still polynomial, as you just need to look it up in E , so it would be $O(|E|)$. Overall creating D would be $O(|E| \cdot |V|^2)$
- k is just $|V|$, so you just need to go through all the nodes, which is linear.

Therefore, overall the function f is computable in polynomial time.

2. **Reliability:** We need to prove $\langle G \rangle \in HAMC \iff f(\langle G \rangle) \in TSP$:

- $\langle G \rangle \in HAMC \rightarrow f(\langle G \rangle) \in TSP$: G has a Hamiltonian cycle, so there is a simple cycle that goes through all the nodes. The edges of this cycle have weight exactly 1 in our TSP instance, and this cycle has exactly $|V| = k$ edges. Therefore, in our TSP instance, there is a tour containing k edges of weight 1, so a tour of overall weight k .
- $f(\langle G \rangle) \in TSP \rightarrow \langle G \rangle \in HAMC$: The constructed TSP instance has a tour going through all vertices of length at most $k = |V|$. Such a tour must also have exactly $|V|$ edges. This means that, by our construction, where edges have length either 1 or ∞ , all the edges of the tour have length 1. These are exactly the edges of the graph G . So the tour only uses edges from E , so it corresponds to a Hamiltonian cycle in G .

- (b) **Given this decision algorithm by the professor and an instance (S, D, k) of TSP, give a polynomial time algorithm that returns an actual TSP-tour through the cities with total cost of at most k , if such a tour exists. If not, the algorithm should return nil.**

FIND-TSP-TOUR(S, D, k)

1. Run TSP-DECIDER(S, D, k)
 - if it rejects, return nil
2. For every unordered pair of distinct vertices (v_i, v_j) :
 - Set $d_{i,j} = d_{j,i} = k + 1$ temporarily
 - Run TSP-DECIDER(S, D, k)
 - if it accepts keep the change
 - if it rejects, revert to the old d values
3. After all pairs useless edges have been processed, the set of edges left with weights under k should encode the tour.

Exercise 3

An important problem in analysing social networks is the question whether there are persons in the network who occur in more than one large clique of the network. Therefore, we consider the following problem:

Name Important Node

Instance an undirected graph $G = (V, E)$, and two positive integers $K > 1$ and $L > 1$.

Question do there exist at least K nodes $v \in V$ such that v occurs in at least two different cliques of G , each containing at least L nodes?

Show that Important Node is an NP problem.

Answer: The following represents a polynomial time verifier for the problem, where the certificate c encodes:

- the K important nodes
- for each of those nodes, two cliques of size at least L that both contain the node

$V =$ "On input $\langle G, K, L, c \rangle$, where

$$c = ((v_1, C_{1,1}, C_{1,2}), \dots, (v_K, C_{K,1}, C_{K,2})),$$

do the following:

1. Check that the certificate contains at least K triples, all with distinct nodes.
2. For each $i \in \{1, \dots, K\}$:
 - (a) Check that $v_i \in V$
 - (b) Check that $|C_{i,1}| \geq L$ and $|C_{i,2}| \geq L$.
 - (c) Check that $C_{i,1}$ and $C_{i,2}$ are both valid cliques.
3. If all checks succeed, accept, otherwise reject."

Exercise 4

Old exam question:

Consider the following problem:

Name ItCanGoBothWays

Instance Given a finite set $S \subseteq \mathbb{N}$ and an integer T

Question Is there a subset $S' \subseteq S$ with $|S'| \leq T$ so that

$$\sum_{s \in S'} s = \sum_{s \notin S'} s?$$

Give an example of a yes-instance and an example of a no-instance of this problem. Ensure that $5 \leq |S| \leq 10$. Include a very brief explanation of why your instances are a yes-instance and a no-instance.

Exercise 5

Old exam question, resit 20–21: Consider the following problem:

Name SatExcept

Instance A set of literals U and a set of clauses C and an integer X .

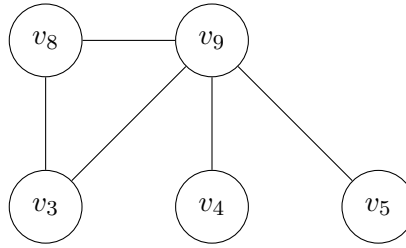
Question Is there a truth assignment satisfying C such that at most X clauses are not satisfied?

Consider the following reduction from an instance $I = (G = (V, E), K)$ of Vertex Cover to SatExcept:

- $U = V$
- $C = \{E \cup \{\{\neg v_i\} \mid v_i \in V\}\}$
- $X = K$

(a) Given the following 2 instances of vertex cover, of which 1 is a YES-instance and 1 is a NO-instance. Show this.

Instance 1: $K = 2$



Instance 2:

$$V = \{v_1, v_2, v_3, v_4\}, \quad E = \{\{v_1, v_2\}, \{v_2, v_3\}, \{v_1, v_4\}\}, \quad K = 1$$

Answer: Instance 1 is a YES-instance, as one of the possible solutions is $\{v_9, v_8\}$. Instance 2 can be shown to be a NO instance by looking at each vertex and seeing if it covers all others. You can then see that there is no solution with $K = 1$ to this instance.

(b) Now pick one of the instances, apply the reduction and show that result is still a yes-instance or no-instance (depending on which one you picked).

$$\begin{aligned} f(I_1) = \langle & \\ & U = \{v_3, v_4, v_5, v_8, v_9\}, \\ & C = \{(v_3 \vee v_8), (v_3 \vee v_9), (v_8 \vee v_9), (v_9 \vee v_4), (v_9 \vee v_5), (\neg v_3), (\neg v_4), (\neg v_5), (\neg v_8), (\neg v_9)\}, \\ & K = 2 \\ & \rangle \end{aligned}$$

Now take the assignment:

$$\{v_3 = 0, v_4 = 0, v_5 = 0, v_8 = 1, v_9 = 1\}$$

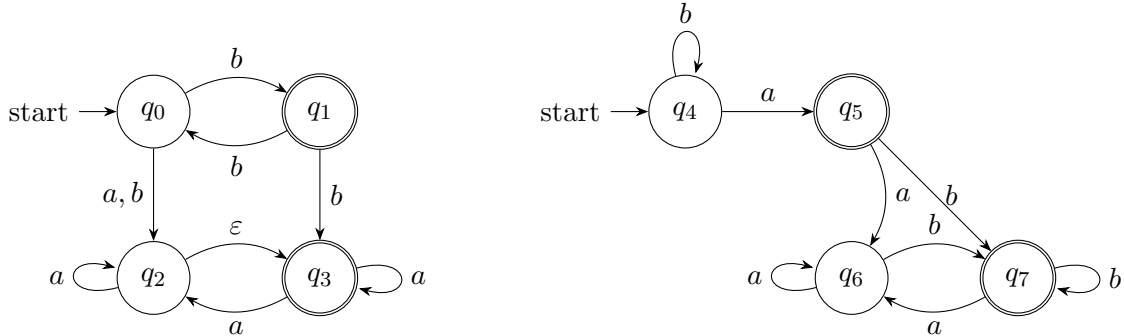
All clauses in C are satisfied except for 2: $(\neg v_8)$ and $(\neg v_9)$. Therefore, this is a YES-instance of SatExcept.

(c) Show that the reduction can be performed in polynomial time.

Stage 1 and 3 are obviously linear, as they just copy parts of the input to the output. Stage 2 first iterates over all edges and copies them over as clauses, which is linear in the number of edges, and then creates a clause for every node, which is obviously linear in the number of nodes. So overall stage 2 is in $O(n + m)$ which is still polynomial in the size of the overall input.

Exercise 6

Revision, old exam question midterm 20–21: Consider an NFA N on the left and an NFA N' on the right with transition diagrams as follows.



Describe how you could use N and N' to construct an NFA N'' such that

$$L(N'') = L(N)^* \circ L(N').$$

(note the ‘star’ symbol after $L(N)$). How would you combine them and what you would need to change? Make sure to explain it for this particular example rather than the general principle.

Exercise 7

Revision, old exam question endterm 21–22: For each of the following claims, show the claim is true or give a counterexample and explanation to show the claim is false, starting your answer with either “True” or “False”.

- (a) If there is a direct reduction from L to A_{TM} and from A_{TM} to L , then $\bar{L}$ is not Turing-recognisable.
- (b) If there is a Karp reduction $SAT \leq_P L$, then L is undecidable.

Bonus Exercise